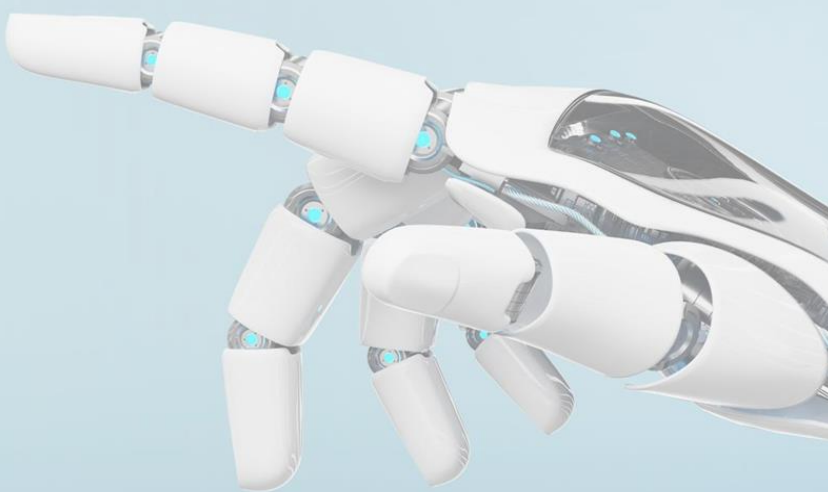




Lilian Laporte



▼ Education

- 09.2022 **M.Sc. in Robotics, Systems & Controls**
05.2025 **ETHZ - Zürich, Switzerland**
Diploma: MSc in Robotics, Systems & Controls (GPA: 5.6/6)
- 09.2019 **B.Sc. in Microtechnics**
07.2022 **EPFL - Lausanne, Switzerland**
Diploma: BSc in Microtechnics (GPA: 5.05/6)
- 08.2021 **Exchange Program - Electrical Engineering, Mathematics and Computer Science**
07.2022 **TUD - Delft, Netherlands**
- 09.2012 **High school and Secondary school**
06.2019 **St. François de Sales - Ville-la-Grand, France** (#1 in France, *Le Monde*, 2015)
Diploma: French Baccalaureate S (mathematical speciality), *with distinction*

▼ Projects

Master's Thesis - Robotics Systems Lab (RSL) - ETHZ

"Guiding Exploration in Fine-Tuning via Simple Trajectory Pre-training for Advanced Skill Learning"

Developed a two-stage learning framework that combines imitation-based pre-training and reinforcement learning for complex robot behaviors, to enable a wheeled quadruped robot to perform stunts (e.g. backflips).

Semester Project - IBM/Acoustic Robotics Systems Lab (ARSL) - ETHZ

"Autonomous Ultrasound-Driven Manipulation of Microrobots with Reinforcement Learning Enhanced by Real-Time ultrasound Imaging Feedback"

Developed a computer vision pipeline for a real-time ultrasound imaging feedback system to control microrobots within the body. Created vein prototypes to demonstrate the concept and control the microrobots.

ARC Robotics Hiwi - ETHZ - Zürich, Switzerland

Assisted the team in improving an automatic waste sorting system by implementing a new computer vision pipeline for waste tracking, developing a simulation of the entire system in Gazebo, and enhancing the presorting stage.

Publication: Esquivel Estay, et al. (2024-04-08). *Dataset and Analysis of River Waste Pollution in the Limmat River, CH, during a One-day City Festival (Zurich Street Parade, 2023)* [White paper].

Xplore Robotics Engineer - EPFL - Lausanne, Switzerland

Responsible for designing and manufacturing the forearm and the gripper of a Mars rover as part of an European inter-university competition.

Awards: - #3 at European Rover Challenge (ERC) for our first year of creation
- Special prize for handling task
- Special prize for science task

CONTACT INFORMATION

Address: Murwiesenstrasse 12, 8057 Zürich, Switzerland
Phone: +33 6 45 88 75 45
Email: llaporte@ethz.ch
Nationality: French
Birth date: 08/07/2001
Portfolio: lilianlaporte.com
Github: [LilianLaporte](https://github.com/LilianLaporte)

TECHNICAL SKILLS

• Programming:

- Python: IsaacLab, mujoco, pytorch, YOLO, OpenCV
- C/C++: GTKmm
- Matlab: yalmip, Simulink, OpenCV
- ROS Noetic
- Docker
- Javascript
- HTML/CSS
- Assembler

• CAD/Simulation:

- Catia
- Solidworks
- Blender
- Abaqus

SOFT SKILLS

- Team communication
- Problem-solving
- Project management
- Creativity
- Ownership

LANGUAGES

- **French** : Native language
- **English** : Level C1
- **German** : Level B2

▼ Work Experience

- 02.2024 **Robotics Intern – MESH AG – Birr, Switzerland (Full-time)**
09.2024
- Developed a robotic system to automate rebar stacking, improving efficiency and accuracy for construction suppliers.
 - Practical experience in Computer Vision, ROS, and robotic software engineering while enhancing the app's functionality and my full-stack skills.
 - **Side project:** RL pipeline in Mujoco to derive an optimal stacking policy based on the shape of the rebar.
- 08.2022 **Machinist – Rolex, Bracelet Dpt – Plan-les-Ouates, Switzerland (Full-time)**
- Managed a 3-axis CNC for manufacturing bracelet stands and polishing them.
 - Firsthand experience in a production chain, understanding key challenges in industrial manufacturing within a multinational company.
- 10.2021 **Lab assistant – AIR Lab – Delft, Netherlands (5h/week)**
05.2022
- Assisted a PhD student in developing code for controlling a robot in the retail supply chain, from warehouses to customers.
 - Initial experience in robotics programming using high-level languages.
- 07.2019 **Electrical Service Assistant – Rolex, Energy Dpt – Plan-les-Ouates, Switzerland (Full-time)**
- Processed career data in the energy sector and finding solutions for electrical issues in a factory of 2000 employees.
 - Hands-on experience with a variety of tasks, enhancing my versatility.
- 11.2015 **Internship – Somfy, R&D Dpt – Cluses, France (Full-time)**
- Explored various engineering sectors within an international firm, from prototyping to production. Gained insight into the importance of communication and collaboration between different departments.

▼ Reference

Raphaël Marwalder – Production Manager – Rolex

Phone number: +41799047757

Ammar Mirjan – CEO – MESH AG

Email address: mirjan@mesh.ch

INTERESTS

- Downhill/Cross-country skiing
- Scuba diving (PADI Open Water)
- Climbing
- Rap/Rnb music